

User Research & Analysis Report

1. Research Goals

The primary goal of this research is to analyze how users interact with wearable health devices and identify the challenges they face in understanding and utilizing health data effectively. The study aims to explore user engagement, pain points, and expectations regarding wearable device interfaces to enhance usability and promote healthier lifestyles.

Research Questions:

- How can wearable health device interfaces be improved to enhance user engagement and promote better health outcomes?
 - What are the key usability challenges users experience with health-tracking devices?
 - How do users currently interact with wearable health data, and what improvements do they desire?
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2. Interview Plan & Schedule

Participants Selection Criteria:

- 5 participants selected based on diverse backgrounds:
 - **1 fitness enthusiasts** who use wearables daily.
 - **3 casual user** who wears a device but does not frequently check data.
 - **1 non-user** who considered purchasing a wearable but decided against it.
- Age range: 22-35 years old
- Devices used: Apple Watch, Fitbit, Garmin...

Interview Schedule:

- **Date:** Conducted from February 12–16, 2025
 - **Format:** 45-60 minute in-person and online (Zoom) interviews
 - **Location:** University campus, coffee shops, remote (Zoom)
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3. Interview Script

Semi-Structured Questions:

A. Background & Usage

1. Do you currently use a wearable health device? If yes, which one and for how long?
2. What features do you use the most? (e.g., heart rate monitoring, sleep tracking, step counting)

B. User Experience & Pain Points

3. How often do you check your device's health data? In what situations do you check it?
4. Do you find the data easy to understand? If not, what confuses you?
5. Have you ever changed your health habits because of the data? If so, how?
6. What challenges have you faced in interpreting or using the data effectively?

C. Improvements & Expectations

7. If you could change anything about the interface or notifications, what would it be?
8. Would you benefit from more personalized recommendations based on your data?
9. Have you ever ignored or disregarded device alerts? Why?
10. If you don't use a wearable device, what prevented you from purchasing

one?

4. Survey Plan

Survey Objective:

To gather quantitative data on user interaction with wearable health devices, usability issues, and feature preferences.

Survey Structure:

- **Target Audience:** Individuals aged 18-40 who currently use or have considered using a wearable health device.
- **Question Types:** Multiple-choice, simple answer, and open-ended questions.
- **Distribution:** Google Forms, shared via email and social media.
- **Number of Responses:** 10-20 collected.

Key Questions:

1. Which wearable devices have you used?
 2. How long have you been using wearable devices for health tracking?
 3. How often do you use the health and fitness tracking features on your wearable device?
 4. Which health metrics do you find most useful on your wearable device?
 5. Do you find the data provided by the wearable device easy to understand?
 6. What challenges do you face when using health tracking features?
 7. What improvements would you suggest for wearable health management devices?
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5. Analysis & Findings

Affinity Diagram Summary:

User responses were categorized into four key themes:

1. Ease of Understanding:

- 3/5 interviewees reported that some metrics (e.g., HRV, stress scores) were hard to interpret.
- 4/5 mentioned that while basic data was clear, actionable insights were lacking.

2. User Engagement & Motivation:

- 2/5 participants actively used wearables for fitness goals.
- 1/5 participant checked health data only when prompted by notifications.

3. Notification Effectiveness:

- 3/5 interviewees found notifications useful but sometimes excessive.
- 2/5 stated they ignored device alerts due to lack of urgency.

4. Desired Improvements:

- Personalized health recommendations (e.g., “Your heart rate has been high today, try a breathing exercise.”)
- Better data visualization with trend analysis.
- More intuitive explanations for complex health metrics.

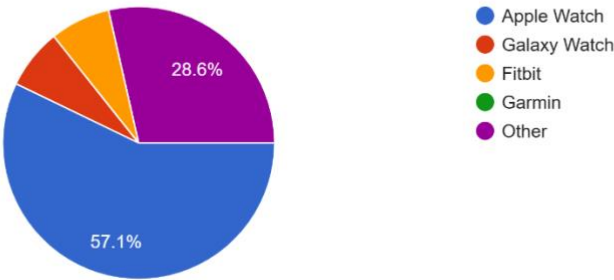
Empathy Map Summary:	
Category	Key Insights
Say	"I don't know what my HRV means, so I ignore it."
Do	Check sleep stats daily but ignore stress tracking.
Think	"Is my resting heart rate normal?"
Feel	Sometimes anxious when seeing unusual readings.

Experience Map Summary:			
Phase	User Action	Pain Points	Opportunities
Wake Up	Checks sleep tracking	Confusing deep sleep data	Provide simpler summaries
Workout	Uses heart rate monitor	No clear training insights	Personalized workout suggestions
End of Day	Reviews daily activity	Overwhelming raw data	Trend analysis recommendations

6. Survey Results (Summarized)

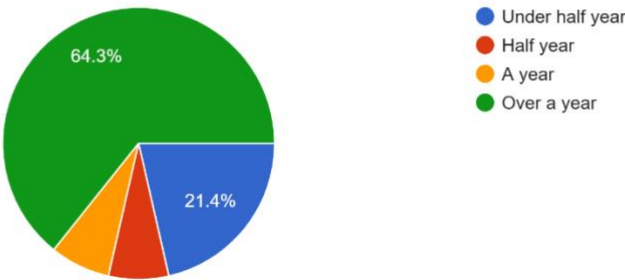
Which wearable devices have you used?

14 則回應

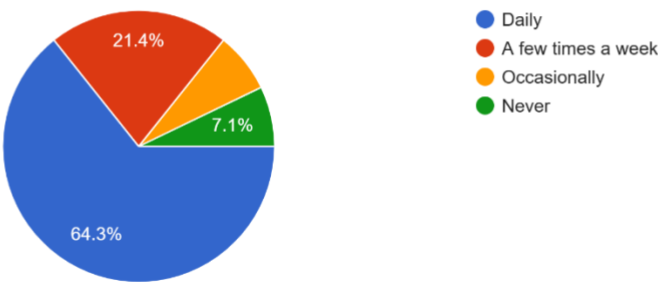


How long have you been using wearable devices for health tracking?

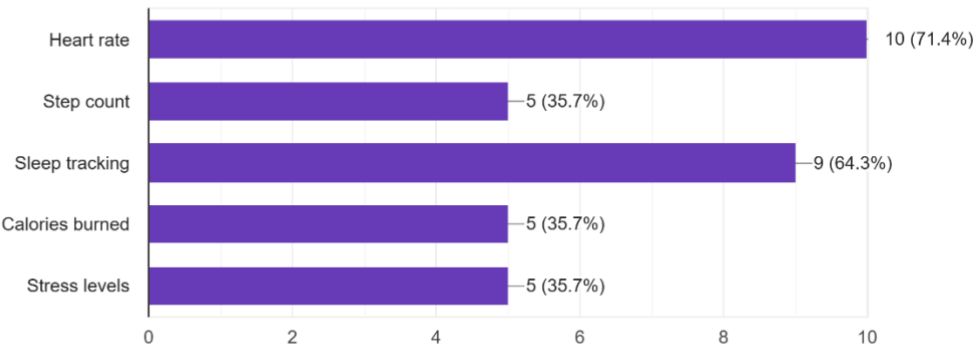
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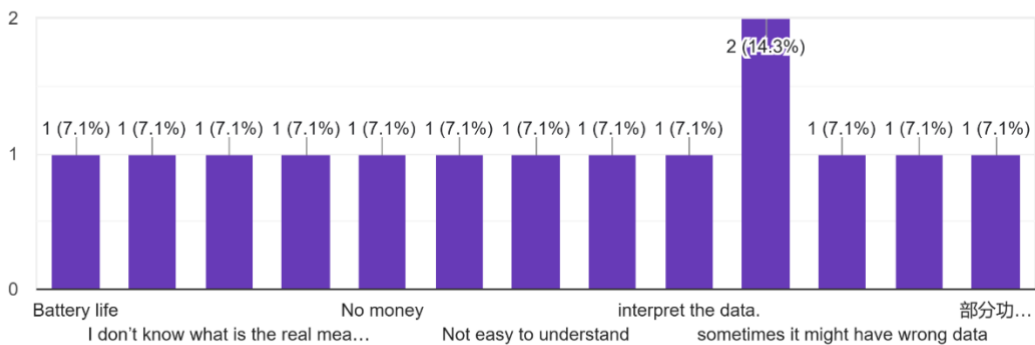
How often do you use the health and fitness tracking features on your wearable device?
14 則回應



Which health metrics do you find most useful on your wearable device?
14 則回應



What challenges do you face when using health tracking features?
14 則回應



7. Learnings from Interview and Survey Data

Key Takeaways:

- Users value **heart rate and sleep tracking** the most.

- Many struggle to **interpret complex data**, leading to disengagement.
 - **Personalized insights and actionable recommendations** would improve user experience.
 - Overuse of notifications leads to **alert fatigue**, reducing responsiveness.
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8. AI Disclosure & Contribution Statement

AI Disclosure:

- ChatGPT was used for structuring interview scripts, summarizing insights, and formatting the final report.
- Grammarly was used for grammar and clarity checks.

Contribution Statement:

- **Hsueh-Ta Lu:** Conducted competitor analysis, summarized user pain points.
 - **Tung-Ying Lee:** Led interviews, analyzed qualitative data, structured findings.
 - **Po-Han Chen:** Designed surveys, compiled quantitative analysis.
 - **Hsin-Ying Tsai:** Created empathy/experience maps, formatted final report.
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Final Remarks

This research provides clear insights into wearable device usability challenges and opportunities for improvement. Future iterations should focus on better **data explanation, personalized insights, and reducing alert fatigue** to enhance user engagement.
